Office Network Infrastructure

Deployment and Documentation Report

Cisco Packet Tracer Simulation

Technologies: VLANs | 802.1Q Trunking | Inter-VLAN Routing | Cisco CLI | TCP/IP

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1. Executive Summary

This report documents the design, configuration, and deployment of a small office network using Cisco Packet Tracer. The project involved building a segmented two-floor network architecture utilizing VLAN technology, 802.1Q trunking, and inter-VLAN routing through a Router-on-a-Stick configuration.

The network was designed to support multiple departments with logical separation, ensuring secure and scalable communication. Throughout the project, hands-on configuration of Cisco switches and routers was performed using the Cisco CLI, along with structured network documentation including port mapping, switch interface documentation, and simulated patch panel connections.

2. Project Overview

2.1 Objectives

- Design and implement a segmented office network with departmental VLANs
- Configure 802.1Q trunk links between switches and from switch to router
- Enable inter-VLAN routing via a Router-on-a-Stick topology
- Perform port mapping and interface documentation for both floors
- Simulate patch panel connections and validate end-to-end IP connectivity
- Troubleshoot and resolve VLAN, trunking, and routing issues

2.2 Network Topology Summary

Component	Details
Simulation Tool	Cisco Packet Tracer
Number of Floors	2 (Floor 1 and Floor 2)
Switches	2 x Cisco Catalyst Layer 2 Switches
Router	1 x Cisco Router (Router-on-a-Stick)
VLANs Configured	VLAN 10 (HR), VLAN 20 (IT), VLAN 30 (Finance) & VLAN 30 (Guest)
Trunk Protocol	IEEE 802.1Q
Routing Method	Inter-VLAN Routing (Router-on-a-Stick)
IP Addressing	Separate subnets per VLAN

2.3 Logical Topology

Traffic flows from end devices through access ports to their respective switch, then across an 802.1Q trunk link to the router's subinterfaces where inter-VLAN routing decisions are made.

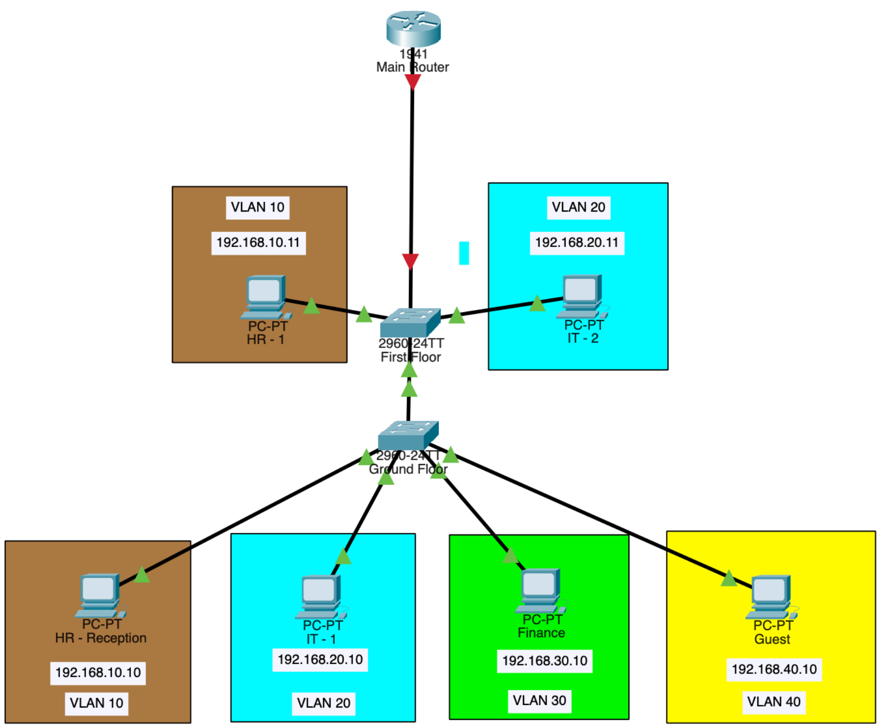

Figure 1 — Full Network Topology in Cisco Packet Tracer

3. VLAN Configuration

3.1 VLAN Design

Virtual Local Area Networks (VLANs) were used to logically segment the office network into isolated broadcast domains per department. This approach enhances security by preventing unauthorized cross-department access at Layer 2, reduces unnecessary broadcast traffic, and simplifies network management.

3.2 VLAN Table

VLAN ID	VLAN Name	Department	Subnet
VLAN 10	HR	Human Resources	192.168.10.0/24
VLAN 20	IT	Information Technology	192.168.20.0/24
VLAN 30	Finance	Management	192.168.30.0/24
VLAN 40	Guest	Guest	192.168.40.0/24

3.3 Switch 1 (Floor 1) - VLAN Creation

VLANs were created on Switch 1 using the Cisco CLI. Each VLAN was assigned a descriptive name and associated with the appropriate access ports connected to end devices on Floor 1.

```
Switch#show vlan brief
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   HR                               active
20   IT                               active
30   FINANCE                          active
40   GUEST                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Figure 2: VLAN Creation on Switch 1 (Floor 1)

3.4 Port Assignment to VLANs

Switch ports were configured as access ports and assigned to their respective VLANs. Access ports carry traffic for a single VLAN and are connected to end-user devices such as workstations and printers.

```
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
Switch(config-if)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
Switch(config-if)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#
Switch(config-if)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#sh
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   HR                               active    Fa0/1
20   IT                               active    Fa0/2
30   FINANCE                          active    Fa0/3
40   GUEST                            active    Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Figure 3: Assigning Ports to VLANs on Switch 1

4. 802.1Q Trunk Configuration

4.1 Overview of Trunking

Trunk links are configured between network devices to carry traffic for multiple VLANs over a single physical connection. The IEEE 802.1Q standard was used, which inserts a 4-byte tag into Ethernet frames to identify the VLAN to which each frame belongs.

4.2 Trunk Link: Switch 1 to Switch 2

A trunk link was established between Switch 1 on Floor 1 and Switch 2 on Floor 2. This allows VLAN traffic to traverse floors without requiring separate physical links per VLAN.

```
Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show trun
Switch#show trunk brief
             ^
% Invalid input detected at '^' marker.

Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/24      10,20,30,40

Port        Vlans allowed and active in management domain
Fa0/24      10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/24      10,20,30,40

Switch#
```

Figure 4: Trunk Link Configuration from Switch 1 to Floor 2

4.3 Trunk Link: Switch 2 to Main Router

Switch 2 on Floor 2 was also configured with a trunk link to the main router. This trunk carries all VLAN traffic to the router, where inter-VLAN routing is performed. The router interface was configured with sub-interfaces corresponding to each VLAN.

```
Floor2#en
Floor2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Floor2(config)#interface fa0/24
Floor2(config-if)#switchport mode trunk
Floor2(config-if)#switchport trunk allowed vlan 10,20,30,40
Floor2(config-if)#
Floor2(config-if)#
Floor2(config-if)#interface fa0/23
Floor2(config-if)#switchport mode trunk
Floor2(config-if)#switchport trunk allowed vlan 10,20,30,40
Floor2(config-if)#
Floor2(config-if)#exit
Floor2(config)#exit
Floor2#
%SYS-5-CONFIG_I: Configured from console by console

Floor2#show interf
Floor2#show interfaces tru
Floor2#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/24      10,20,30,40

Port        Vlans allowed and active in management domain
Fa0/24      10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/24      10,20,30,40

Floor2#
```

Figure 5: Switch 2 Trunk Configuration Toward Router

5. Port Mapping and Interface Documentation

5.1 Switch 1 (Floor 1) - Port Map

The following table documents all active switch interfaces on Switch 1, including their assigned VLANs, interface modes, and connected devices. This documentation serves as a reference for network audits and future configuration changes.

Port	Mode	VLAN	Connected Device	Notes
Fa0/1	Access	VLAN 10 (HR)	HR-PC1	End device
Fa0/2	Access	VLAN 20 (IT)	IT-PC1	End device
Fa0/3	Access	VLAN 30 (Finance)	FIN-PC1	End device
Fa0/4	Access	VLAN 40 (Guest)	GUEST-PC1	End device
Fa0/24	Trunk	VLANs 10,20,30,40	SW2 Fa0/24	802.1Q uplink

5.2 Switch 2 (Floor 2) - Port Map

Port	Mode	VLAN	Connected Device	Notes
Fa0/1	Access	VLAN 10 (HR)	HR-PC2	End device
Fa0/2	Access	VLAN 20 (IT)	ENG-PC2	End device
Fa0/23	Trunk	VLANs 10,20,30,40	Router G0/0	Uplink to R1
Fa0/24	Trunk	VLANs 10,20,30,40	SW1 Fa0/24	Inter-switch

6. Inter-VLAN Routing Configuration

6.1 Router-on-a-Stick Architecture

Inter-VLAN routing was implemented using a Router-on-a-Stick topology. A single physical interface on the main router was divided into logical sub-interfaces, each associated with a specific VLAN. The 802.1Q encapsulation on each sub-interface allows the router to identify and route traffic between VLANs.

This method is efficient for small office deployments as it requires only one physical link between the switch and the router, while still supporting full inter-VLAN communication.

6.2 Router Sub-Interface Configuration

Subinterface	VLAN	IP Address	Subnet Mask	Gateway For
G0/0.10	VLAN 10 (HR)	192.168.10.1	255.255.255.0	HR PCs
G0/0.20	VLAN 20 (IT)	192.168.20.1	255.255.255.0	Eng PCs
G0/0.30	VLAN 30 (Finance)	192.168.30.1	255.255.255.0	Finance PCs
G0/0.40	VLAN 40 (Guest)	192.168.40.1	255.255.255.0	Guest PCs

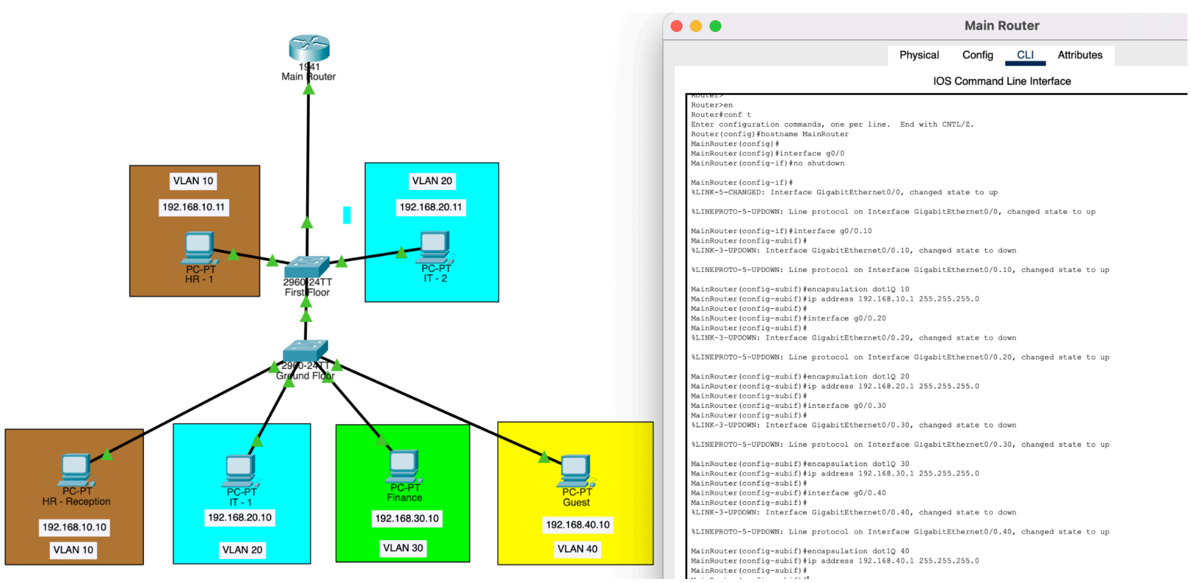

Figure 6: Router Subinterface Configuration (dot1Q Encapsulation)

7. Patch Panel and Physical Layer Documentation

7.1 Simulated Patch Panel Overview

Patch panel connections were simulated to represent the physical cabling infrastructure of the office. Patch panels serve as the central termination point for horizontal cabling runs from workstations to the network closet, where they are then patched to switch ports using short patch cables.

7.2 Floor 1 Patch Panel Table

Device	VLAN	IP Address	Subnet Mask	Default Gateway
HR-PC1	VLAN 10	192.168.10.10	255.255.255.0	192.168.10.1
HR-PC2	VLAN 10	192.168.10.11	255.255.255.0	192.168.10.1
IT-PC1	VLAN 20	192.168.20.10	255.255.255.0	192.168.20.1
IT-PC2	VLAN 20	192.168.20.11	255.255.255.0	192.168.20.1
FIN-PC1	VLAN 30	192.168.30.10	255.255.255.0	192.168.30.1
GUEST-PC1	VLAN 40	192.168.40.10	255.255.255.0	192.168.40.1

8. Troubleshooting and Issue Resolution

8.1 Issue 1 - VLAN Not Propagating Across Trunk

Symptom

Devices on VLAN 20 (IT) on Floor 2 could not communicate with devices on VLAN 20 on Floor 1 after the trunk was established.

Root Cause

The trunk link on Switch 1 was configured with a restricted allowed VLAN list that did not include VLAN 20.

Resolution

The allowed VLAN list on the trunk interface was updated using the command: switchport trunk allowed vlan add 20. This restored cross-floor VLAN connectivity.

8.2 Issue 2 - Router Interface Showing as Down

Symptom

After trunking traffic from Switch 2 on Floor 2 to the main router, the router interface link remained red (down) in Packet Tracer.

Root Cause

The physical router interface had not been brought up using the no shutdown command. By default, Cisco router interfaces are administratively down.

Resolution

The no shutdown command was applied to the physical interface, and subsequently to each sub-interface. The interface came up and inter-VLAN routing became functional.

8.3 Issue 3 - IP Connectivity Failure Between VLANs

Symptom

Pings between devices on VLAN 10 and VLAN 20 failed even after the router sub-interfaces were configured.

Root Cause

The default gateway on end devices was not configured. Without a default gateway pointing to the router sub-interface IP, hosts could not send traffic destined for other VLANs.

Resolution

Default gateways were assigned to each host in their respective VLAN (e.g., 192.168.10.1 for VLAN 10 hosts). Subsequent ping tests confirmed successful inter-VLAN connectivity.

9. Testing and Validation

9.1 Same-VLAN Connectivity Test

Connectivity within the same VLAN was verified first. Devices on the same VLAN successfully pinged each other, confirming that access port assignments and VLAN creation were correct.

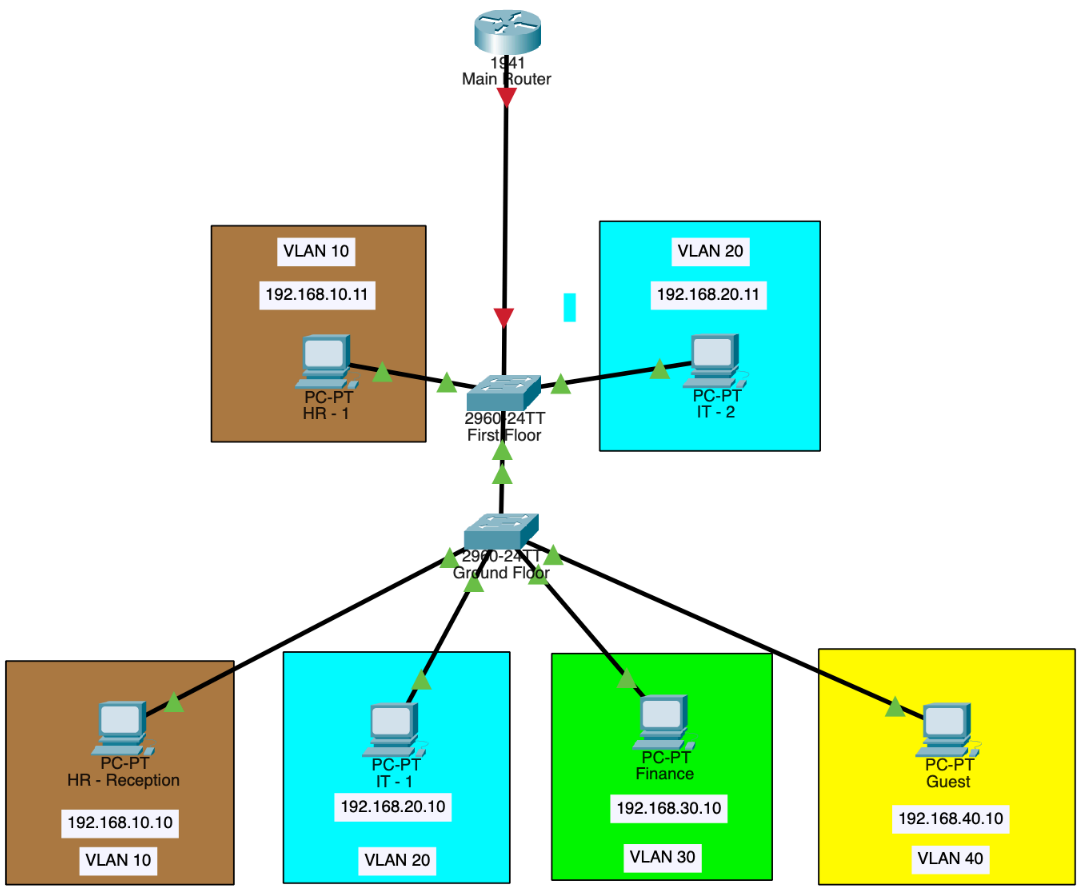

Figure 7: Same-VLAN Ping Test (Successful)

9.2 Inter-VLAN Connectivity Test

After the Router-on-a-Stick configuration was completed and default gateways were applied to all endpoints, inter-VLAN ping tests were executed. Devices in VLAN 10 successfully communicated with devices in VLAN 20 and VLAN 30, confirming that routing was functioning correctly.

10. Conclusion

This project successfully demonstrated the design, configuration, and validation of a segmented two-floor office network using Cisco Packet Tracer. The deployment covered the full lifecycle of a small enterprise network, from initial VLAN design and switch configuration to inter-switch trunking, inter-VLAN routing, and systematic troubleshooting.

Key outcomes achieved throughout the project include:

- Successful VLAN segmentation across two floors using IEEE 802.1Q trunking
- Functional inter-VLAN routing implemented via a Router-on-a-Stick topology
- Comprehensive port mapping and interface documentation for both switches
- Simulated patch panel cabling documentation aligned with physical cabling best practices
- Identification and resolution of three distinct network issues related to VLAN propagation, interface state, and IP gateway configuration
- Validated end-to-end connectivity for both same-VLAN and cross-VLAN traffic

The skills applied in this project reflect core competencies expected of a network technician or junior network engineer, including Switch Configuration, VLANs, Port Mapping, Network Documentation, Patching, Troubleshooting, Cisco CLI, and TCP/IP.